

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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ABSTRACT

Knowledge-intensive reasoning systems now expose intermediate traces, but a reviewer under a fixed inspection budget still needs to know which elements to inspect first and, more importantly, why. Two steps with the same salience can demand opposite review actions: one may consolidate a redundant cluster, another may gate downstream reasoning and must be protected. Flat step scores collapse this distinction. This paper presents Structurally-Calibrated Functional Attribution (SC-FMA), a method that turns a supplied utility or proxy-fidelity signal into an audit queue where each selected step carries decomposed fidelity, graph-necessity, redundancy, and bottleneck fields that record the reason it is prioritized. Its Structurally-Calibrated Utility (SCU) objective solves a convex weighting problem over process steps: it preserves the informative input signal while adding graph-aware audit structure. We evaluate SC-FMA through an Evidence Ladder spanning a controlled synthetic stage, a Countries-KG typed-edge stage, PRM800K process-supervision annotations (4,417 samples and 34,219 labeled steps under a frozen hash split), and oracle audit-target validation. SC-FMA preserves ranking quality while turning a scalar ordering into an inspectable rationale: on rule-derived audit-target validation, the decomposition lifts Recall@25% from 0.235 to 0.699 and NDCG@25% from 0.353 to 0.978 over a scalar-only view, and a Sentence-BERT ablation confirms the graph layer accepts a semantic-embedding backend. SC-FMA thus establishes a signal-preserving audit-prioritization layer for observable reasoning traces: it preserves ranking accuracy and, for each queued step, exposes whether to inspect local evidence, consolidate redundant checks, protect a structural bottleneck, or repair an upstream signal.

1. Introduction

Knowledge-intensive reasoning systems increasingly leave traces that can be inspected after a prediction is made. A rule fires, an entity path is checked, a retrieval result is used, or a reflective language-agent step critiques a partial answer. Step-level supervision has shown that intermediate verification can reveal errors hidden by final-answer accuracy [1, 2]. Retrieval-augmented and knowledge-graph-augmented pipelines add a related opportunity: retrieved passages [3] and entity-relation structure [4] are available as review objects. The practical bottleneck is no longer whether traces exist, but which trace elements deserve attention under a limited audit budget.

A flat salience score can order trace elements, but it rarely tells a reviewer how to act on that order. A high-ranked step may be locally reliable, structurally necessary, redundant with neighboring evidence, or a bottleneck for downstream reasoning. These distinctions matter in KBS settings such as rule chains, retrieval-augmented workflows, and KG-assisted reasoning, where inspection time is limited and different structural roles imply different review actions.

This paper studies audit prioritization as intervention-based functional attribution over observable reasoning traces. We propose Structurally-Calibrated Functional Attribution (SC-FMA), a structural calibration layer that preserves an informative utility or proxy-fidelity signal while adding graph necessity, redundancy, and bottleneck diagnostics. The Structurally-Calibrated Utility (SCU) objective implements the calibration as a convex weighting problem over process steps. The resulting weight is both a rank and an audit record: it identifies which step enters the queue and why a reviewer should inspect it.

The main findings are summarized as follows:

- On the controlled synthetic stage, SC-FMA QP exhibits the intended calibration behavior in a high-structure-conflict setting, increasing mean Spearman correlation with proxy step labels from 0.483 for raw CIU to 0.597.

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A component ablation shows that the fidelity anchor is the essential signal, while redundancy and bottleneck terms act as conditional regularizers whose contribution scales with structural preference in the labels.

- On PRM800K process-supervision annotations (4,417 samples and 34,219 labeled steps under a frozen hash split), a learned fidelity signal is the primary direct ranker (Spearman $\rho = 0.611$). SC-FMA Ridge closely preserves it ($\rho = 0.604$) while exposing decomposable priority rules for audit.
- An automatic audit-target validation provides rule-derived (non-human) evidence that SC-FMA decomposition retrieves bottleneck, redundancy, weak-anchor, and structural-overcorrection targets more effectively than a scalar-only view. Mean Recall@25% increases from 0.235 to 0.699, and mean NDCG@25% increases from 0.353 to 0.978.
- A Sentence-BERT graph-construction ablation and the fixture-level Countries-KG typed-edge stage show that the graph layer can accept a semantic-embedding backend and typed ontology edges at the same interface.

The paper makes three contributions. First, it formulates fixed-budget review of reasoning traces as an audit-prioritization problem. Second, it introduces SC-FMA and the SCU objective, which convert fidelity or utility inputs into structurally calibrated verification-step weights with formal guarantees and decomposable audit fields. Third, it evaluates the method through an Evidence Ladder that moves from controlled synthetic calibration to Countries-KG typed-edge feasibility, PRM800K step-label ranking, and rule-derived audit-target validation. Across this ladder, SC-FMA preserves useful ranking signal while exposing the structural reasons behind the priority order. Section 4.6 states the evidence boundary and future validation directions.

The remainder of this paper is organized as follows. Section 2 reviews attribution, process supervision, and graph-based KBS audit methods, and identifies the research gap. Section 3 presents the SC-FMA methodology, the SCU objective, and its theoretical properties. Section 4 applies SC-FMA along the Evidence Ladder and reports the audit readout. Section 5 concludes.

Figure 1 summarizes the audit workflow. SC-FMA converts a flat process score into a decomposed decision over an observable trace. The trace is represented as an audit graph, the SCU objective combines fidelity and structural diagnostics, and each selected step is reported through an Audit Card. The result is a fixed-budget review queue with an inspectable rationale for each selected step.

2. Related Work

2.1. Attribution and Explanation

Attribution methods ask which parts of an input, computation, or structure matter for a decision. Integrated Gradients attributes a prediction to input features along a path from a baseline [5]; SHAP unifies additive feature-attribution methods [6]; and LIME explains individual predictions through local interpretable surrogates [7]. Rationale-oriented work such as ERASER evaluates extractive spans rather than only continuous scores [8].

Structure-aware attribution adds another layer. Network studies show that importance can be non-local, redundant, or concentrated around bottleneck nodes [9, 10]. For graph models, GNNExplainer selects compact subgraphs that explain predictions [11]. User-centered explanation work further shows that an explanation is useful only when it matches the reviewer's task [12].

These lines motivate SC-FMA but leave a gap for audit triage. Saliency identifies important evidence, yet it does not say whether the reviewer should inspect local support, consolidate duplicated evidence, protect a downstream dependency, or repair a weak upstream signal. Entailment-tree methods move explanation toward intermediate justifications [13], and counterfactual methods move it toward contrastive evidence [14]. SC-FMA keeps these audit roles separate instead of folding them into one score.

2.2. Process Supervision and Process Reward Models

Process supervision scores intermediate steps rather than only final answers. PRM800K showed that process reward models trained on per-step human labels can outperform outcome reward models in verifier-guided best-of- N search [1]. The GSM8K verifier setup made this training route concrete [15], and process-based feedback reduced reasoning errors even when outcome accuracy appeared saturated [2]. These studies provide the kind of step-resolved fidelity signal that SC-FMA can calibrate.

Structurally-Calibrated Functional Attribution

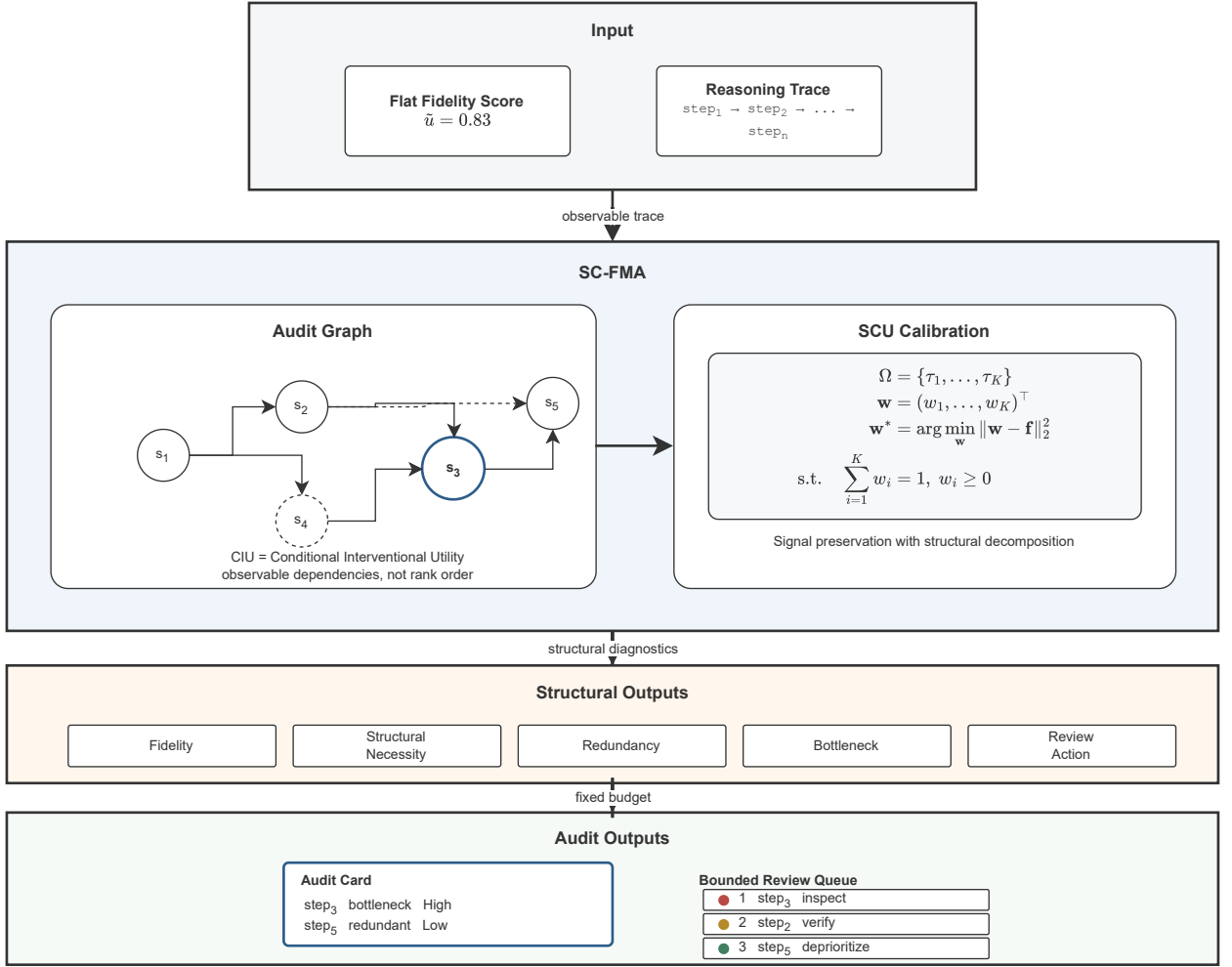


Figure 1: Overall SC-FMA audit-prioritization framework. The figure shows how an Input (输入) consisting of a flat fidelity score and an observable reasoning trace is transformed by SC-FMA (结构校准功能归因) into an audit representation rather than a simple step ranking. The Audit Graph (审计图) encodes structural dependencies among trace steps, with CIU denoting Conditional Interventional Utility. Structural Outputs (结构化输出) expose fidelity, structural necessity, redundancy, bottleneck status, and review action. The Audit Outputs (审计输出) consist of an Audit Card (审计卡) and a Bounded Review Queue (有界审查队列). Arrows indicate the data flow: observable trace information is converted into graph diagnostics, calibrated by the SCU objective, decomposed into interpretable review factors, and finally constrained into a fixed-budget inspection queue.

A parallel line makes verification, critique, and revision themselves available as scoring units. Chain-of-thought prompting elicits intermediate reasoning [16], and self-consistency aggregates multiple sampled chains into a more reliable signal [17]. Tree of Thoughts extends search from linear chains to deliberate tree-structured exploration [18], while ReAct interleaves reasoning and acting for verification [19]. On the revision side, Reflexion [20] and Self-Refine [21] introduced reflective agents that critique and revise their own outputs, turning revision steps into an additional supervision target.

Recent benchmarks also show that step-level signals need distributional checks. Math-Shepherd studies automatic step verification without human annotations [22], and ProcessBench evaluates process-error identification across models [23]. These resources help assess scalar step scores, but audit prioritization needs more than a scalar. SC-FMA therefore treats process-supervision outputs as inputs to structural calibration rather than as the final explanation.

2.3. Graph-Based and KBS Audit Methods

Graph-based and KBS methods show how structured intermediate states can be made inspectable. Classic systems treated auditability as a design principle: MYCIN exposed certainty-factor chains that connected conclusions to rules [24], and DENDRAL exposed generate-and-test plans for chemical structure enumeration [25]. Cognitive architectures such as Soar and ACT-R also made internal states and operator applications available for step-level review [26, 27].

Modern KBS research shifts the audit object from hand-built rules to learned pipelines that connect language models with structured knowledge. RAG exposes retrieved passages as evidence [3], and survey work documents the broader family of RAG variants [28]. GraphRAG builds entity graphs and pre-generates community summaries for global queries [4]. KG-RAG connects a pre-existing knowledge graph to LLM generation through chain-of-exploration reasoning [29]. Knowledge-enhanced language models and LLM-KG integration further expose entity links, triples, relation paths, and graph-quality signals as audit objects [30–34]. Similar audit pressure appears in long-tail KG-augmented question answering and engineering-design RAG, where sparse factual support and heterogeneous evidence make exhaustive review unrealistic [35–37].

Across these settings, the challenge is not simply recovering intermediate evidence. Modern pipelines already expose many passages, entity links, and relation paths. The harder question is how a reviewer should allocate a limited curation budget across them. SC-FMA connects to the KBS audit tradition by converting observable traces into graph-aware audit records that preserve a fidelity signal while making each selected step’s structural role explicit.

2.4. Summary of Research Gap

Existing work therefore supplies three ingredients: local attribution identifies salient evidence, process supervision supplies step-level fidelity, and graph/KBS methods expose structural context. SC-FMA combines these ingredients into an inspectable priority rule for fixed-budget review.

3. Methods

3.1. Formal Problem Definition

Input. SC-FMA takes an observable reasoning trace $T = (s_1, \dots, s_k)$, where each step s_i is a verification, reflection, retrieval, or rule-like operation available for audit. The trace is converted into a graph $G = (V, E)$ whose nodes correspond to steps and whose edges encode temporal adjacency plus optional topical or domain-specific relations. The method also receives a utility or fidelity signal u over the same steps. This signal may come from local utility estimation, a proxy score, or a learned step-ranking model, and SC-FMA treats it as the evidence anchor to be preserved when it is informative.

Output and objective. The output is a calibrated weight vector $\mathbf{w}$ over trace steps. Sorting by $\mathbf{w}$ produces an audit-prioritized ranking, while the associated fidelity, graph-necessity, redundancy, and bottleneck fields explain why each step enters the queue. The objective is to preserve useful utility signal, incorporate observable trace structure, and attach each selected step to a concrete review action.

3.2. SCU Objective and Audit Decomposition

For a trace with k reflective or verification steps, let $\tilde{\mathbf{c}} \in \mathbb{R}^k$ denote the normalized utility or proxy-fidelity input. In the PRM800K locked stage, $\mathbf{w}_{\text{struct}}$ denotes the frozen Ridge regressor used as this fidelity signal. In the binary-correctness CIU setting used by the diagnostic pipeline, $\tilde{\mathbf{c}}$ is a coarse trace-level utility anchor: it records how a trace-level outcome changes under a documented intervention. Let $\tilde{\mathbf{n}}$ denote normalized structural necessity from graph perturbation diagnostics, $\mathbf{R}$ a redundancy matrix, and $\mathbf{b}$ a bottleneck indicator.

SC-FMA computes weights $\mathbf{w}$ on the probability simplex

$$\mathcal{W} = \{\mathbf{w} \in \mathbb{R}_+^k : \sum_i w_i = 1\} \quad (1)$$

by minimizing the SCU objective:

$$L(\mathbf{w}) = \alpha \|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta \|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma \mathbf{w}^\top \mathbf{R} \mathbf{w} - \delta \sum_i b_i \log(w_i). \quad (2)$$

The four terms encode fidelity to the supplied input, structural alignment, redundancy regularization, and non-zero bottleneck regularization. The fidelity anchor consistently provides the dominant optimization signal. The redundancy

penalty and bottleneck log-barrier mainly act as regularization terms that encourage interpretable and decomposable audit reasoning, although their quantitative contribution to ranking metrics is modest under the current evaluation setting. Detailed notation, construction rules, algorithms, and proofs are provided in the supplementary material.

3.3. Trace Representation

Each trace is represented as an ordered set of verification or reflection steps. The current implementation supports spans derived from structured reflection tags and free-text chain-of-thought segmentation; in this paper, segmentation is treated as an observed input rather than a solved NLP problem. A node v_i corresponds to one step. Temporal edges connect adjacent steps, while topical edges connect steps whose TF-IDF similarity exceeds the configured threshold. This temporal-plus-topical graph is the default SCU instantiation for text-only traces and supplies auditable structural decomposition. A supplementary graph-construction ablation replaces TF-IDF topical edges with Sentence-BERT embeddings (sentence-transformers/all-MiniLM-L6-v2, 384 dimensions): in that 800-trace fixture, the TF-IDF topical default (0.0515) did not outperform temporal-only edges (0.0596) on diagnostic structural-necessity Spearman, while the embedding backend (0.0615) slightly improved the same diagnostic score. These numbers are too small for the topical edges to be read as a ranking driver, so we interpret the graph layer as an interchangeable *construction interface* rather than as evidence that the TF-IDF topology adds ranking value over the temporal chain. The graph constructor is intentionally modular: richer domain-specific backends — semantic embeddings, entity links, knowledge graphs, ontologies, rule-dependency DAGs, and data-lineage graphs — can replace the default without modifying the SCU objective, as illustrated by the Countries-KG typed-edge stage in the Evidence Ladder (Section 4.3) and the embedding ablation. The resulting graph is a directed acyclic approximation of an observable reasoning chain and serves as a reproducible audit graph.

3.4. Structural Necessity, Redundancy, and Bottlenecks

Structural necessity measures how much graph support is lost when a node is perturbed under a documented graph-level operation. The implementation uses PRUNE, CASCADE, and BYPASS diagnostics. PRUNE removes direct node support. CASCADE propagates disruption to downstream neighbors. BYPASS asks whether alternative paths can substitute for the removed step. The final necessity input $\tilde{\mathbf{n}}$ is the normalized summary of these observable graph diagnostics.

The redundancy matrix $\mathbf{R}$ captures pairwise similarity between steps. High redundancy means that two steps appear to serve overlapping functions in the observed trace. The penalty $\mathbf{w}^T \mathbf{R} \mathbf{w}$ discourages assigning high weight to many mutually redundant steps at once. In the current evidence package, this term is best interpreted as a regularization and decomposition term: it records a possible duplicated-audit allocation pressure, but its direct contribution to ranking metrics is modest.

Bottleneck detection addresses the opposite diagnostic pattern. Some nodes are not redundant, and their downstream exposure can make them useful to inspect even if the supplied fidelity signal is modest. The log-barrier term prevents identified bottleneck nodes from being assigned near-zero weight. In KBS terms, this is analogous to keeping a single point of failure visible to the reviewer. The term enforces a conservative audit regularizer when observed structure marks the node as difficult to replace.

Figure 2 illustrates the CIU-necessity mismatch. Figure 3 compares the three perturbation modes, and Figure 4 shows the resilience curves.

Theorem 3.1 (SCU well-posedness). *If either the quadratic condition*

$$\alpha + \beta > 0 \quad \text{and} \quad \mathbf{R} \succeq 0, \quad (3a)$$

or the log-barrier condition

$$\delta > 0 \quad \text{for at least one node with } b_i = 1 \quad (3b)$$

holds, then the SCU objective in Eq. 2 is convex on the simplex in Eq. 1. Under the quadratic condition in Eq. 3a, positive regularization on the feasible subspace gives a unique minimizer; under the log-barrier condition in Eq. 3b, an active barrier keeps identified bottleneck nodes strictly inside the feasible region. For non-redundant step pairs, the calibrated weights preserve the joint ordering induced by $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$ under the stated monotonicity condition; bottleneck nodes with $b_i = 1$ receive strictly positive weight when Eq. 3b applies.

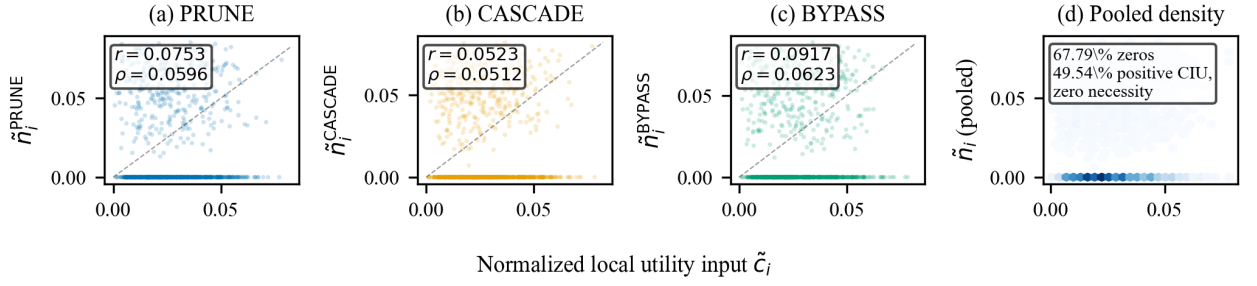


Figure 2: Structural diagnostic motivating calibration. Local utility and graph-level necessity are not interchangeable: many locally useful steps have low measured structural necessity, while some structurally exposed steps do not receive strong local utility signals. The shared x-axis $\tilde{c}_i$ denotes the normalized local CIU utility input for step i . SC-FMA uses this mismatch as the reason for calibration.

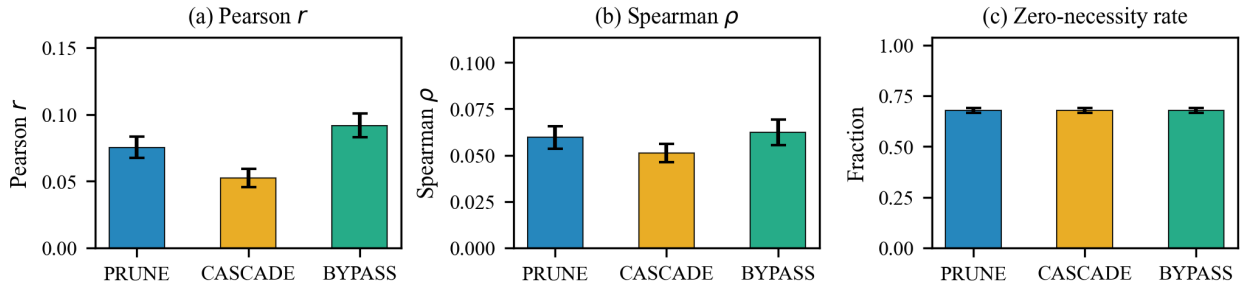


Figure 3: Structural diagnostic summary by perturbation mode. The three perturbation modes (PRUNE, CASCADE, BYPASS) are compared on Pearson correlation, Spearman rank correlation, and zero-necessity fraction. Error bars indicate 95% bootstrap confidence intervals. The consistency across modes indicates that weak CIU-necessity alignment is not an artifact of a single perturbation strategy.

We implement three variants as implementations of the same calibration framework, not as competing claims that a single optimizer is universally best. SC-FMA Ridge uses a closed-form softmax over a linear combination of $\tilde{c}$ and $\tilde{n}$. SC-FMA QP solves the full constrained program with redundancy and bottleneck terms. SC-FMA Projection applies a fast topology-constrained simplex projection onto the probability simplex [38]. The evaluation sections report where each variant is appropriate rather than treating any one optimizer as universal.

The calibrated weights are intended to be inspectable rather than opaque. For each selected step, the implementation can report whether the score was driven mainly by fidelity, graph necessity, redundancy regularization, or bottleneck regularization. This decomposition is important for KBS use because different actions follow from different reasons. A high-fidelity step may be a strong candidate for positive process supervision. A high-necessity step may be a candidate for manual review even when its local utility estimate is modest. A highly redundant group may indicate that several checks can be audited together rather than separately. A bottleneck step may deserve conservative retention in the review queue because later reasoning depends on it.

This decomposition also prevents a common interpretation error. SC-FMA does not say that the largest weight is the only important step in a trace. The simplex constraint represents a review budget, so increasing one step's weight necessarily decreases another step's share of attention. The output is therefore best read as a priority allocation under a fixed budget. When two steps are redundant, assigning both high weight is usually wasteful; when a step is isolated and downstream-exposed, assigning it near-zero weight can hide a fragile part of the trace. The SCU objective encodes these two pressures explicitly.

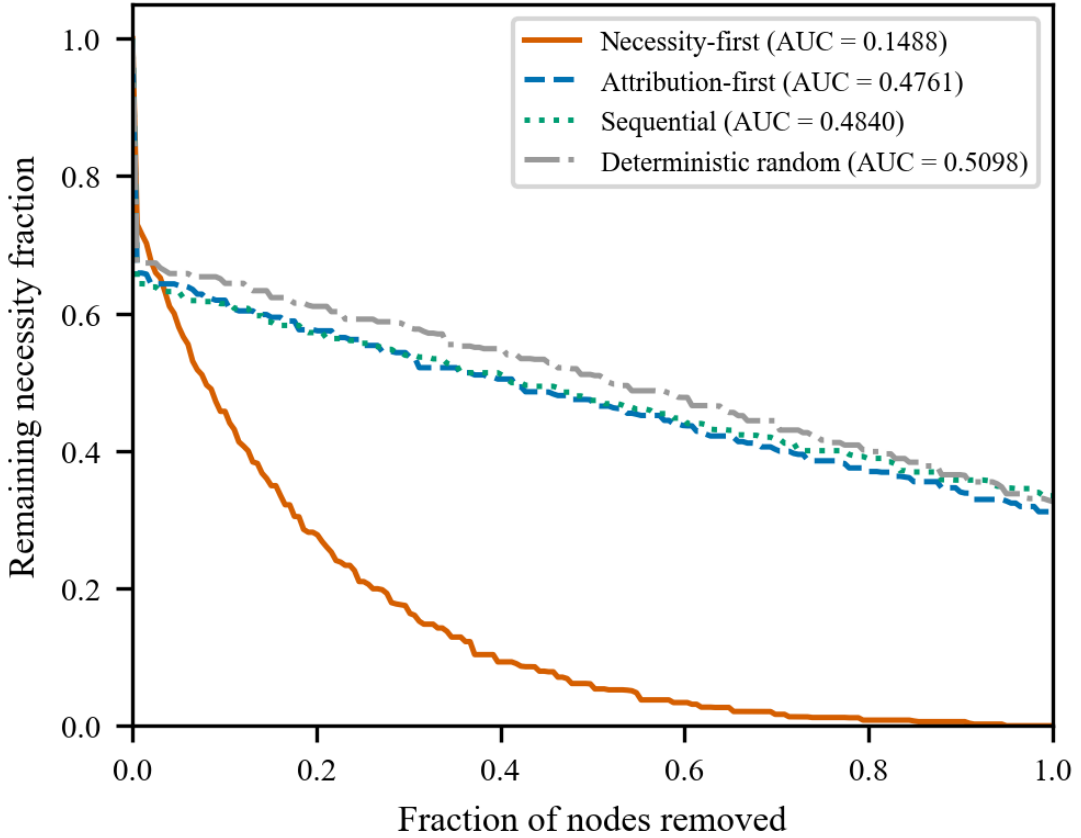


Figure 4: Resilience curves under ordered node removal. Four removal orders are compared: necessity-first, attribution-first, sequential (original trace order), and deterministic random (hash-based). The y-axis shows remaining structural necessity fraction; the x-axis is the fraction of nodes removed. AUC values are annotated. The necessity-first curve degrades sharply (AUC 0.1488), while attribution-first (AUC 0.4761) is close to random (AUC 0.5098), showing that structural criticality requires graph-level diagnostics beyond local CIU.

3.5. Role of Structural Components

The structural components are not designed to outperform a strong fidelity baseline in every setting. Their role is to preserve useful ordering when the fidelity signal is strong. They also add audit information that the scalar alone cannot expose: whether a step is locally well supported, structurally exposed, redundant with nearby checks, or protected as a bottleneck.

A fidelity-only scalar answers the ranking question but collapses the rationale for that ranking. A reviewer cannot distinguish whether a high-ranked step should be followed as local evidence, inspected as a downstream gatekeeper, or consolidated with a redundant group. These distinctions imply different audit actions. SC-FMA therefore changes the interpretability of the priority order even when the order itself changes only modestly: the contribution is preservation with decomposition.

3.6. Computational Profile

Empirical runtime is reported separately from the asymptotic summaries. In the frozen local artifacts, the PRM800K audit-prioritization run over 4,417 traces and 34,219 steps completed in 131.57 s. Separately logged graph-construction, SCU component, and failure-taxonomy runs are reported in Appendix B.

Table 1

Asymptotic time complexity for SC-FMA fixed-budget audit. Here N is the number of traces, k is the number of steps in a trace, L is text length, d is the sparse text-feature dimension, and I is the number of QP solver iterations.

Stage	Input size	Time complexity
Trace feature extraction	N traces, k steps, text length L	$O(NkL)$
Graph construction	k nodes, sparse TF-IDF features d	$O(k + k^2d)$
Redundancy and bottleneck diagnostics	k steps, similarity matrix	$O(k^2d)$
Ridge calibration	k step feature rows	$O(k)$ after learned coefficients
QP calibration	k steps, dense constraints	$O(Ik^3)$ worst case
Fixed-budget ranking	k calibrated weights	$O(k \log k)$

Table 2

Memory profile for the same pipeline stages. Dense graph quantities are shown for clarity; the implementation uses sparse storage where applicable.

Stage	Memory complexity
Trace feature extraction	$O(k + d)$ per trace
Graph construction	$O(k^2)$ dense; sparse in implementation
Redundancy and bottleneck diagnostics	$O(k^2)$
Ridge calibration	$O(k)$
QP calibration	$O(k^2)$
Fixed-budget ranking	$O(k)$

4. Application

This section evaluates SC-FMA as an Evidence Ladder rather than as a single aggregate score. Each stage follows the workflow in Figure 1: convert observable reasoning traces into audit graphs, calibrate a fidelity anchor against graph diagnostics, and report both a fixed-budget queue and an Audit Card for selected steps. The controlled synthetic stage isolates calibration behavior under designed structure conflict. The Countries-KG typed-edge stage checks whether the same graph interface can accept typed domain structure. The PRM800K locked stage then tests ranking preservation and fixed-budget audit prioritization on a larger process-annotation distribution. Finally, the oracle audit-target validation stage asks whether decomposition fields recover audit targets that scalar scores hide. Section 4.1 fixes the data, baselines, and Evidence Ladder accounting for the later stages; Section 4.6 then interprets the audit decomposition, states the variant-selection policy, and gives the evidence boundaries.

4.1. Experimental Setup

4.1.1. Data and Baselines

The controlled synthetic stage contains 200 traces generated with five fixed seeds (42, 123, 456, 789, 1024), with approximately 1,000 reflective steps per seed. It is used for method development, calibration, ablation, and error analysis. Controlled proxy step labels combine local correctness, lexical diversity, position, and graph-derived necessity. The PRM800K locked stage uses process-supervision annotations under the frozen v3.6/v3.8 hash split (4,417 samples and 34,219 labeled steps). The hash split fixes sample assignment before locked evaluation to avoid post hoc selection. MuSiQue is reported as a supplementary constructed-label feasibility demonstration because its labels and scoring features share `step_type`; full metrics and execution details are reported in the supplementary material. WebQSP trace-audit artifacts are reported as a supplementary fixed-schema diagnostic on executable traces (494 train traces for development and 1,627 locked test traces). GSM8K/HotpotQA task-specific replay and downstream filtering attempts are retained as failed or incomplete validation attempts.

Computed baselines include raw CIU, span length, relative position, random ranking, and stage-specific scalar or feature baselines. Token-level attribution families such as gradient attribution, attention rollout, Monte Carlo Shapley, token surprisal, and step entropy are not computed under the archived workflow because the synthetic artifacts store

Table 3

Evidence Ladder accounting. Each stage has a separate manuscript interpretation.

Evidence Ladder stage	Primary artifact	Allowed interpretation
Controlled synthetic stage	200 traces, approximately 1,000 steps, five fixed seeds	Controlled calibration and ablation evidence.
Countries-KG typed-edge stage	Fixture-level Countries-KG diagnostic	Typed-edge implementation feasibility.
PRM800K locked stage	4,417 samples, 34,219 labeled steps	PRM800K-like step-label ranking and audit context.
Oracle audit-target validation stage	Rule-derived PRM800K audit targets	Automatic audit-target information recovery.
Frozen PRM comparison	Prefix-sequence reward scores	In-distribution baseline context.
Supplementary MuSiQue/WebQSP diagnostics	Supplementary details only	Feasibility and fixed-schema diagnostic context only.
Failed GSM8K/HotpotQA attempts	Failed or blocked artifacts	Transparency only; no positive validation claim.

per-step CIU, necessity, and redundancy rather than token-level activations; they are therefore discussed only under the reproducibility scope. The primary metric is Spearman rank correlation; Kendall τ , NDCG@3, NDCG@5, and fixed-budget audit-prioritization metrics are reported where relevant. Hyperparameter grids, including the gamma/delta and alpha/beta QP sensitivity grids (Supplementary Tables C.5 and C.6), additional ablations, error modes, and efficiency measurements are reported in the supplementary material.

The controlled synthetic stage is designed to test calibration behavior rather than to mimic all properties of natural reasoning. CIU values are skewed, necessity is zero-inflated, and redundancy is block structured. These choices reflect the diagnostic pattern that motivated SC-FMA: local utility can be dense, while structural necessity is sparse. The controlled proxy label is deliberately not identical to any SCU input. It combines correctness, lexical diversity, position, and structural information so that a method cannot win by copying one input vector. Supplementary Table D.12 reports cross-correlations between proxy-label components and SCU inputs to document this separation.

For the larger real-data stage, PRM800K is treated as a process-annotation distribution. The locked v3.6 split evaluates how well step-weighting signals rank annotated steps. The v3.8 frozen PRM comparison provides in-distribution baseline context using prefix-sequence reward probabilities. Because PRM800K can overlap with public PRM training distributions, Section 4.6 handles the scope of PRM-training and best-of- N extensions.

All reported comparisons are deterministic under the archived configs. The controlled synthetic stage uses five fixed seeds (42, 123, 456, 789, 1024) with variance reported as $\pm$ standard deviation. Supplementary Table B.1 reports bootstrap standard errors by trace size, Supplementary Table C.8 reports PRM800K confidence intervals for the main calibration variants, and Supplementary Table C.9 reports the extended fixed-budget audit readout. Extended Holm-Bonferroni, Friedman/Nemenyi, paired Wilcoxon, and effect-size analyses are retained in the supplementary reproducibility notes, while Section 4.6 centralizes Evidence Ladder boundaries.

4.1.2. Evidence Ladder and Claim Accounting

The evaluation is organized as an Evidence Ladder rather than by a single aggregate score. Table 3 is the canonical accounting device: each row states what a stage can support and what it cannot support. This prevents controlled synthetic calibration, typed-edge feasibility, PRM800K step-label ranking, oracle audit-target validation, supplementary diagnostics, and failed replay attempts from being merged into a stronger claim than any individual stage permits.

The distinction also affects how null or negative results are interpreted. The PRM800K variant comparison shows that the full QP can be too aggressive when the base fidelity signal already aligns well with annotation ordering. The contribution is therefore not a single leaderboard number; it is a reproducible way to combine local and structural evidence while keeping each Evidence Ladder stage tied to its data-generating conditions.

Table 4

Step importance ranking on the controlled synthetic stage (200 traces, approximately 1,000 steps). For SC-FMA variants, the Spearman ρ column is the 5-seed mean $\pm$ standard deviation (seeds 42, 123, 456, 789, 1024) and the remaining columns are from a representative seed (42), since the archived multiseed run records Spearman alone. All other methods are from seed 42. Simple average is an independent baseline computed as the element-wise mean of normalized CIU and structural necessity. Token-level attribution baselines (gradient, attention-rollout, Shapley, surprisal, entropy) are not reported here because the archived synthetic pipeline stores only per-step CIU/necessity/redundancy rather than token-level activations, so those methods would collapse to raw CIU; they are discussed in the supplementary material under the reproducibility scope. Bootstrap confidence intervals and extended ablations are reported in the supplementary material.

Method	Spearman ρ	Kendall τ	NDCG@3	NDCG@5
SC-FMA QP	0.597 $\pm$ 0.064	0.533	0.921	0.958
SC-FMA Ridge	0.541 $\pm$ 0.016	0.447	0.889	0.927
Raw CIU	0.483	0.412	0.874	0.919
Simple average (CIU + necessity)	0.478	0.413	0.889	0.926
SC-FMA Projection	0.372 $\pm$ 0.030	0.278	0.824	0.890
Span Length	-0.006	-0.013	0.740	0.826
Relative Position	-0.005	-0.004	0.735	0.827
Random	-0.010	-0.008	0.736	0.825

Table 5

SCU component contribution on the controlled synthetic stage. Values are 5-seed mean $\pm$ standard deviation over seeds 42, 123, 456, 789, and 1024. Delta is Spearman difference relative to full_scu. The largest drop occurs when the fidelity anchor is removed; removing redundancy or bottleneck regularization has near-zero impact on ranking metrics in this setting.

Variant	Spearman ρ mean $\pm$ std	Kendall τ mean $\pm$ std	Delta vs Full
full_scu	0.5103 $\pm$ 0.0320	0.4297 $\pm$ 0.0287	0.0000
no_fidelity	0.2850 $\pm$ 0.0153	0.2366 $\pm$ 0.0138	-0.2253
no_structure	0.4841 $\pm$ 0.0334	0.4053 $\pm$ 0.0285	-0.0263
no_redundancy	0.5093 $\pm$ 0.0321	0.4273 $\pm$ 0.0273	-0.0010
no_bottleneck	0.5090 $\pm$ 0.0304	0.4279 $\pm$ 0.0274	-0.0013

4.2. Synthetic Step-Importance Ranking

Table 4 shows the intended calibration behavior on the controlled synthetic stage: SC-FMA QP aligns more closely with proxy step labels than raw CIU, with a Spearman ρ gap of 0.114 between the QP five-seed mean (0.597) and the raw-CIU representative-seed value (0.483). The result supports the methodological claim that structural calibration can correct local-utility weights when the stage provides controlled proxy step labels. The next step in the Evidence Ladder asks whether this graph-calibration interface can move beyond text-only synthetic structure and accept typed domain relations from a real knowledge graph.

Table 5 refines the earlier component narrative. The fidelity anchor is the essential anchor in this experiment: removing it reduces Spearman by 0.2253, far more than any other removal. Graph necessity contributes moderately (Delta = -0.0263). By contrast, removing the redundancy penalty or bottleneck log-barrier changes Spearman by only about 0.001. These terms should therefore be described as regularization and decomposition terms rather than as ranking drivers under the current evaluation. This component-ablation table reports full_scu Spearman 0.5103, which is lower than the SC-FMA QP value of 0.597 in Table 4. The two tables use different controlled synthetic configurations (the component-ablation run versus the multi-seed ranking run), so their absolute values are not directly comparable. The within-table ablation deltas are the meaningful quantity here, not the cross-table absolute level.

The near-zero redundancy and bottleneck deltas on the original controlled synthetic stage reflect a property of its labels, not of the terms. The proxy label there is a random vector independent of redundancy and bottleneck structure, so the structural terms have nothing to align to. To test the regime they are designed for, we run a structural stress-test (Table 6) whose labels explicitly encode structural preference. Steps participating in any redundant pair are demoted

Table 6

SCU component contribution on a structural stress-test benchmark (200 traces per seed, five fixed seeds 42, 123, 456, 789, 1024). Unlike the structure-independent labels of Table 5, these labels reward redundancy-awareness and bottleneck-protection under a stress-test redundancy setting ($\gamma = 0.6$). Delta is the Spearman difference relative to `full_scu`.

Variant	Spearman ρ mean $\pm$ std	Kendall τ mean $\pm$ std	Delta vs Full
<code>full_scu</code>	0.6249 $\pm$ 0.0145	0.5238 $\pm$ 0.0112	0.0000
<code>no_fidelity</code>	0.4832 $\pm$ 0.0357	0.3935 $\pm$ 0.0319	-0.1417
<code>no_structure</code>	0.5871 $\pm$ 0.0234	0.4870 $\pm$ 0.0224	-0.0378
<code>no_redundancy</code>	0.5758 $\pm$ 0.0163	0.4889 $\pm$ 0.0091	-0.0492
<code>no_bottleneck</code>	0.5409 $\pm$ 0.0145	0.4424 $\pm$ 0.0114	-0.0840

Table 7

Graph construction analysis on the diagnostic run. Temporal edges define the primary backbone; topical similarity edges are supplementary cues. NDCG@3 and NDCG@5 are computed from the same diagnostic-run source data and scoring definition used for the correlation readout.

Variant	Spearman ρ	Kendall τ	NDCG@3	NDCG@5
<code>full_tfidf</code>	0.0515	0.0458	0.8655	0.8655
<code>temporal_only</code>	0.0596	0.0527	0.8688	0.8688
<code>jaccard_topical</code>	0.0517	0.0460	0.8657	0.8657
<code>shuffled_topical</code>	0.0515	0.0458	0.8655	0.8655

in the label (redundancy-awareness), and bottleneck steps are boosted and given a weak local utility (bottleneck-protection). The fidelity input blends this adjusted label with the pre-adjustment base label. It therefore remains the dominant anchor but is partially structurally blind, leaving the structural terms corrective work.

On the stress-test, removing the fidelity anchor still costs the most (-0.1417), indicating that it remains the dominant anchor. The structural terms now contribute materially: removing the bottleneck log-barrier costs 0.0840, the redundancy penalty 0.0492, and graph necessity 0.0378, all far larger than the ≈ 0.001 observed for redundancy and bottleneck on the structure-independent synthetic setting. The appropriate interpretation is that the redundancy and bottleneck terms are *conditional* regularizers: their contribution scales with how much structural preference the fidelity input fails to encode. This is the intended KBS audit setting, where a locally plausible step can duplicate another check (redundancy) or a rare check can carry a weak local signal while gating downstream inference (bottleneck). On stages whose labels are structure-independent they reduce to near-no-op regularizers, which is why Table 5 reports $\Delta \approx 0.001$ for them.

Table 7 shows that topical edges do not add ranking-preservation evidence in the diagnostic run: `full_tfidf` is slightly below `temporal_only` and matches `shuffled_topical` on the reported ranking metrics. The appropriate interpretation is therefore conservative. Topical edges can change structural diagnostics such as mean necessity, bridge-node fraction, and influence depth, but their observed role is supplementary structural cueing rather than a demonstrated ranking driver.

The pattern across baselines is informative. Raw CIU already ranks moderately above the heuristic baselines (span length, relative position, random), confirming that the supplied utility signal carries useful ordering. The independent simple-average of CIU and structural necessity is comparable to raw CIU, so plain averaging of the two anchors does not add ranking value; the calibrated SC-FMA weightings (QP and Ridge) are needed to lift ranking above raw CIU. Token-level attribution baselines (gradient, attention rollout, Shapley, surprisal, entropy) are not reported in this table because the archived synthetic pipeline stores per-step CIU/necessity/redundancy rather than token-level activations, so those methods would collapse to raw CIU under the archived workflow; we therefore frame SC-FMA only against the baselines actually computed here and leave token-level attribution to a live-model evaluation. The QP result supports the value of explicitly modeling graph structure in the controlled synthetic stage.

The component and graph-construction analyses also separate two claims that could otherwise be conflated. The SCU objective benefits primarily from fidelity anchoring and secondarily from graph-necessity alignment in this stage. Redundancy and bottleneck terms mainly make the audit reasoning decomposable and conservative. Separately,

Table 8

Countries-KG ontology-aware edge construction stage (30 traces over a real knowledge graph, 30 entities, 189 triples). KG-augmented edges are derived from typed ontology relations.

Quantity	Value
KG entities / triples	30 / 189
KG relation types	locatedIn, neighborOf
Reasoning traces	30
Baseline (TF-IDF) edges	310
KG candidate edges	304
KG-augmented edges	320
KG-derived edges added	196
Ontology relations detected	depends_on_concept (211), same_constraint (85), neighbor_concept (8)
Mean abs structural-necessity delta	0.431
Structural-necessity Spearman (baseline vs KG)	0.0
Bottleneck top-1 overlap	1.000
Matched nodes	154

topical graph edges move structural diagnostics but do not yield a ranking gain over temporal-only or shuffled-topical construction. This supports using graph construction as a diagnostic representation, not as an independently validated performance driver.

Error analysis identifies three main residual patterns. First, high-utility but zero-necessity steps can be overweighted because the fidelity term still matters. Second, bottleneck nodes near the detector threshold can be underweighted when the indicator is not activated. Third, dense redundancy blocks can over-equalize weights when many steps share similar topical features. These failure modes are useful because they indicate where future work should focus: finer-grained utility signals, more robust bottleneck detection, and typed KBS edges that distinguish true functional duplication from surface similarity.

4.3. Countries-KG Typed-Edge Stage

The controlled synthetic stage shows when structural calibration can alter a priority order, but it does not show whether the graph interface can ingest domain relations. The Countries-KG typed-edge stage addresses that narrower question. As a diagnostic check on non-lexical graph construction, we use the Countries KG [39], which has 30 entities (countries, regions) and 189 triples over the typed relations `locatedIn` and `neighborOf`. We generate 30 deterministic reasoning traces over KG queries (location lookup, neighbor verification, region connectivity) and compare a TF-IDF-only baseline reflection graph against a KG-augmented graph that maps KG ontology relations to functional edge types. This stage supports implementation feasibility for typed graph construction on real KG structure.

Table 8 shows that typed KG edges change the graph substantially: the stage considers 304 KG candidate relations, adds 196 ontology-derived edges, and yields 320 KG-augmented graph edges versus 310 TF-IDF baseline edges. The mean absolute structural-necessity delta is 0.431 with a baseline-vs-KG necessity Spearman of 0.0, meaning KG augmentation substantially *reorders* which steps the method flags as structurally necessary rather than merely adding redundant connections. This is the KBS-relevant capability the methodology targets: prioritizing steps via domain structure (typed ontology relations) rather than lexical overlap alone. Because this stage is fixture-level graph-construction evidence only, the next Evidence Ladder stage turns to PRM800K for larger locked process-annotation evidence. Full edge lists and the diagnostic report are in the supplementary material.

4.4. PRM800K Evidence and Audit Readout

The locked PRM800K result is specific to the third Evidence Ladder stage. After the Countries-KG stage establishes typed-edge feasibility at fixture scale, PRM800K supplies a larger process-annotation distribution for ranking preservation and fixed-budget audit prioritization. The stored $\mathbf{w}_{\text{struct}}$ ranking is the primary positive signal, with Spearman $\rho = 0.611$ against PRM800K step ratings. Raw local utility is negative on this stage ($\rho = -0.077$), and the overlap-limited frozen PRM prefix-score baseline is lower ($\rho = 0.252$). SC-FMA Ridge closely preserves $\mathbf{w}_{\text{struct}}$

Table 9

Merged PRM800K real-data ranking and fixed-budget audit-prioritization readout on the frozen split of 4,417 samples and 34,219 labeled steps. Metrics are displayed to three decimals from archived JSON artifacts. Dashes for the Simple average row indicate that Top-1 Hit and Mass@25% were not computed for this negatively correlated diagnostic baseline.

Signal	Spearman ρ	Top-1 Hit	Mass@25%	NDCG@25%	Role
$\mathbf{w}_{\text{struct}}$	0.611	0.917	0.381	0.951	Primary learned fidelity signal.
SC-FMA Ridge	0.604	0.905	0.380	0.946	Preservation-with-decomposition approximation.
SC-FMA QP	0.442	0.903	0.379	0.944	Audit context on this stage.
Frozen PRM prefix score	0.252	0.777	0.342	0.847	In-distribution baseline context.
Raw local utility	-0.077	0.686	0.282	0.722	Direct ablation.
Simple average (CIU + necessity)	-0.443	—	—	0.438	Independent feature-average baseline.
Random	0.006	0.733	0.307	0.776	Seeded sanity control.

($\rho = 0.604$), while QP ($\rho = 0.442$) and Projection ($\rho = -0.135$) are lower-scoring variants. Supplementary Tables C.8 and C.9 report the extended confidence intervals and fixed-budget readouts.

The merged ranking and fixed-budget readout in Table 9 shows that, on PRM800K-like process annotations, $\mathbf{w}_{\text{struct}}$ and Ridge place more high-rated steps inside the review prefix than raw local utility and simple controls. The simple-average baseline (unweighted mean of CIU and structural necessity, Spearman $\rho = -0.443$) performs poorly on this stage, indicating that learned weighting is needed to recover annotation ordering.

The PRM800K variant pattern is informative. The $\mathbf{w}_{\text{struct}}$ signal defined in Section 3.2 reaches Spearman $\rho = 0.611$, showing that the fidelity input is already strong on the locked split. QP’s hard structural constraints are designed for settings where utility and structure disagree; when fidelity already aligns with annotation ordering, those constraints can reallocate weight away from the stronger signal. The Discussion returns to why calibration remains useful beyond direct ranking (Section 4.6.4).

For fixed-budget audit, the key question is whether a reviewer sees higher-rated process steps earlier. NDCG@25%, top-1 hit, and mass@25% match this review setting more directly than whole-trace rank correlation alone. In a KBS maintenance workflow, a curator may inspect only a small number of rule firings, retrieval checks, entity bindings, or reflective verification steps from each trace. The value of the queue is therefore practical: decide what to inspect first and record the reason for that priority.

4.5. PRM800K Error Case Analysis

The stratified audit-prioritization results (reported in full in the supplementary material) reveal systematic patterns in where SC-FMA variants succeed and where they degrade on PRM800K data. This analysis is diagnostic: it characterizes the conditions under which structural calibration helps or hinders step ranking, rather than constructing a one-number ranking claim.

A failure taxonomy further converts these diagnostics into variant selection guidance. The taxonomy is multi-label over 4,417 locked PRM800K traces. Weak utility anchor appears in 54.79% of traces, structural over-correction in 48.56%, redundancy misclassification in 27.48%, low signal or tie in 18.32%, and bottleneck over-protection in 8.65%. The high weak-utility-anchor rate reinforces the need to preserve the learned $\mathbf{w}_{\text{struct}}$ fidelity signal. The structural-over-correction rate explains why full QP should not be treated as the default PRM800K variant. Redundancy and bottleneck cases appear often enough to discuss directly, but the component study in Table 5 shows that their current quantitative contribution to aggregate ranking metrics is modest. Fixed-template Audit Cards are provided in Appendix A.

4.5.1. Trace Length and Redundancy Density

The strongest predictor of variant behavior is trace length. On short traces (7,141 steps across 1,806 samples), $\mathbf{w}_{\text{struct}}$ achieves Spearman $\rho = 0.759$, SC-FMA Ridge preserves it at 0.757, and SC-FMA QP remains close at 0.734. On long traces (18,814 steps, 1,416 samples), $\mathbf{w}_{\text{struct}}$ drops to 0.447, Ridge to 0.430, and QP degrades more sharply to 0.172. This pattern is consistent with the hypothesis that longer traces contain more redundancy. Step pairs with overlapping topical features create dense blocks in the redundancy matrix. When redundancy density is high, QP's hard redundancy penalty forces weight reallocations that deviate from the annotation signal. Ridge's soft averaging avoids this overcorrection, which explains its greater robustness across trace-length strata.

4.5.2. Label Entropy and Annotation Structure

Stratification by label entropy reveals a complementary pattern. On low-entropy traces (17,288 steps, 1,474 samples), annotation labels are concentrated and discriminability is limited. All methods show compressed metrics, but $\mathbf{w}_{\text{struct}}$ and Ridge still achieve NDCG@25% above 0.99. On high-entropy traces (5,983 steps, 1,304 samples), labels are more dispersed. QP ($\rho = 0.642$) performs relatively closer to $\mathbf{w}_{\text{struct}}$ ($\rho = 0.705$) than on other strata, suggesting that explicit redundancy management becomes more useful when annotation structure is richer and step-level discriminability matters more.

4.5.3. Variant Behavior Summary

Across all six strata (trace length low/mid/high $\times$ label entropy low/mid/high, plus error-uncertainty present/absent), a consistent ordering emerges. $\mathbf{w}_{\text{struct}}$ leads, Ridge preserves closely (within 0.01–0.02 Spearman ρ across most strata), and QP degrades on high-redundancy or long-trace conditions. The frozen PRM prefix score provides moderate in-distribution baseline context (Spearman 0.061–0.425 across strata) and remains consistently above raw local utility. The Projection variant is consistently the weakest SC-FMA variant, with negative or near-zero Spearman values on most strata. These patterns are variant selection guidance: QP is appropriate for controlled settings with explicit redundancy structure, while Ridge is the recommended variant when the fidelity input already aligns well with annotation ordering, as it does on PRM800K via the learned $\mathbf{w}_{\text{struct}}$ signal.

4.5.4. Key Diagnostic Finding

The core takeaway for future SC-FMA use is that QP underperforms when redundancy density is high and $\mathbf{w}_{\text{struct}}$ already captures annotation ordering through its learned feature representation. In that setting, the hard redundancy penalty overcorrects. Ridge is more robust across strata because it soft-averages structural information into the fidelity signal without forcing reallocations. This finding supports a simple methodological recommendation: use Ridge when a strong fidelity baseline exists, reserve QP for settings where utility and structure disagree and redundancy management is needed, and avoid Projection when the structural signal can conflict with fidelity inputs. This variant-level diagnostic is part of the contribution because it turns a simple leaderboard into an actionable design guideline.

4.6. Audit Interpretation and Boundaries

4.6.1. KBS Implications

SC-FMA is relevant to KBS methodology because it separates four audit questions that are often conflated: whether a step is locally useful, structurally necessary, redundant, or a bottleneck. In rule-chain, retrieval-augmented, or KG-assisted workflows, high-necessity bottleneck checks naturally map to review priority, while highly redundant checks may be consolidation candidates. This design principle connects SC-FMA to the KBS tradition of auditable reasoning chains [24, 26]. A KBS-facing use of SC-FMA would proceed in four stages: trace capture, graph construction with optional typed KBS edges, structural diagnostics, and calibration. Here we evaluate that idea as audit prioritization, not as a deployed KBS intervention.

Knowledge maintenance naturally turns this design principle into an audit-prioritization problem. Knowledge-intensive systems accumulate stale triples, noisy retrieved supports, repeated evidence paths, and fragile bottleneck relations; exhaustive inspection is unrealistic when curator time is fixed. Recent KBS work illustrates the same pressure from different angles: KG prompting studies show that LLM use of structured knowledge is sensitive to format, organization, noise, and incompleteness [34]; engineering-design RAG turns explicit domain facts into retrieval objects whose quality affects downstream synthesis [36]; and KG quality evaluation work treats triple quality, feature redundancy, and low-dimensional efficiency as practical maintenance concerns [33]. SC-FMA does not solve these maintenance tasks directly. It supplies the audit layer that such tasks require: under a fixed budget, it distinguishes

evidence to inspect, redundant checks to consolidate, dependency-exposed steps to protect, and weak scoring anchors to repair.

4.6.2. Limitations and Claim Boundary

The evidence supports a methodological contribution with clear scope. The main positive result is preservation with decomposition: Ridge preserves a strong learned fidelity signal while exposing fidelity, necessity, redundancy, and bottleneck pressure. For KBS-facing review, this turns a ranked queue into concrete actions: inspect, consolidate, protect, or repair intermediate reasoning steps under a fixed budget.

Limitation 1 (Controlled synthetic stage): The synthetic calibration stage is controlled and proxy-labeled (200 traces, approximately 1,000 steps per seed, with the main five-seed configuration using seeds 42, 123, 456, 789, and 1024). It supports calibration behavior and ablation analysis under designed conditions. The proxy labels combine correctness, lexical diversity, position, and structural information to test calibration, not to mimic all natural annotations.

Limitation 2 (stage-specific data bounds): The PRM800K locked stage supports real step-label ranking and fixed-budget audit within a PRM800K-like annotation distribution. The locked v3.6 hash split (4,417 samples, 34,219 steps) provides reproducible real-data context. Claims about other task distributions require separate validation. SC-FMA Ridge is reported as preserving the learned signal while adding decomposition fields, rather than exceeding the learned w_{struct} baseline.

Limitation 3 (Graph construction): Current graph construction uses temporal order and topical similarity (TF-IDF cosine) as default edges. Formal ontology semantics, typed rule dependencies, database lineage, and domain-specific relation types are not used in the primary evaluation stages. A supplementary graph-construction ablation replaces TF-IDF with Sentence-BERT embeddings and records the model, version, dimension, cache path, and runtime status. The embedding comparison slightly improves diagnostic structural alignment, and the Countries-KG typed-edge stage in Section 4.3 shows how typed edges can be constructed.

The current TF-IDF graph is a lightweight default implementation; the SCU objective is compatible with richer structures such as semantic embeddings, entity links, ontology edges, rule dependencies, and data lineage without changing the calibration framework.

Limitation 4 (Utility signal granularity): The binary-correctness CIU signal used by the diagnostic pipeline is a coarse trace-level utility anchor. It records how a trace-level outcome changes under a documented intervention. The SCU objective treats it as one input among several, so the resulting weights are local functional attributions rather than causal effect estimates.

The current KBS-facing evidence is limited to audit prioritization and does not validate a deployed KBS workflow.

Excluded claims. Four uses remain outside the current evidence:

- **Downstream PRM training:** SC-FMA has not been evaluated as a training signal for process reward models, reinforcement learning from process feedback, or best-of- N search.
- **Production KBS validation:** The evaluation does not include a live knowledge-based system with domain experts, gold-standard curation decisions, typed ontology edges, downstream error reduction, or maintenance-cost measurement.
- **Causal identification:** The framework estimates local functional attribution, not population-level treatment effects or counterfactual outcomes under unobserved confounding.
- **External PRM generalization:** The frozen PRM baseline comparison (v3.8) provides in-distribution baseline context, with PRM800K overlap risk for public PRM training distributions.

Stage-specific details, including GSM8K/HotpotQA replay failures, WebQSP KGQA boundaries, and QP-specific caveats, are documented in the supplementary material and appendices.

4.6.3. Scope and Applicability

SC-FMA is applicable when the goal is audit prioritization, process diagnosis, or fixed-budget review of observable reasoning or verification traces. In these settings, a ranked step is accompanied by fidelity, graph necessity, redundancy, and bottleneck fields that indicate what a reviewer should inspect.

Table 10

Oracle-based automatic audit-target validation on the locked PRM800K split. Failure-mode localization is evaluated as retrieval under a 25% budget. The SC-FMA condition uses decomposition fields; the scalar condition uses only $\mathbf{w}_{\text{struct}}$.

View	Recall@25%	NDCG@25%	MRR	Top-1 Hit	Cost Proxy
$\mathbf{w}_{\text{struct}}$ scalar only	0.235	0.353	0.354	0.000	0.0007
SC-FMA decomposition	0.699	0.978	1.000	1.000	0.0002

Training-signal use, causal-inference use, production deployment claims, and human-performance claims require different validation designs. The oracle-based validation introduced below tests information recovery from automatic rule-derived audit targets; it does not measure reviewer speed or accuracy. Future deployment studies should use domain-specific traces, typed edges, adjudicated curation targets, domain-expert review, and maintenance-cost or error-reduction measurements.

Within the present evidence, the strongest practical guidance is variant selection. Ridge is robust when a learned fidelity signal already aligns with annotations, whereas QP should be reserved for settings where structural conflict is expected and evaluated directly. The Audit Cards in Appendix A make this guidance operational by showing how the four decomposition fields convert a ranking disagreement into a concrete reviewer action.

4.6.4. Audit Interpretation

Direct ranking answers “which step comes first?” but not “what should the reviewer check?” A $\mathbf{w}_{\text{struct}}$ score gives an effective ordering; SC-FMA adds whether the priority comes from local fidelity, graph necessity, redundancy pressure, or bottleneck protection. Those components turn the same ranked step into different audit actions: inspect local evidence, consolidate a repeated group, protect a dependency, or repair an upstream utility signal.

The merged fixed-budget readout in Table 9 should be interpreted in that light. $\mathbf{w}_{\text{struct}}$ remains the strongest direct queue under the locked PRM800K hash split, and SC-FMA Ridge closely preserves it (NDCG@25% of 0.946 versus 0.951) while retaining the component-level explanation. The comparison against random and raw local utility confirms that informed feature-based ranking is stronger than uninformed baselines; the negative raw-local-utility result reported in Table 9 also explains why the audit queue should begin from a stronger fidelity signal and then expose structural rationale.

The PRM800K locked stage tests whether the ranking signal can be preserved, but it does not by itself test whether the added decomposition fields expose information hidden by a scalar queue. The oracle audit-target validation stage addresses this final step of the Evidence Ladder. We evaluate failure-mode localization as an information-retrieval task over traces. The scalar condition receives only the $\mathbf{w}_{\text{struct}}$ score summary, while the SC-FMA condition receives the Audit Card fields: fidelity, necessity, redundancy, bottleneck status, and induced action. Automatic oracle targets are derived from frozen PRM800K labels, ranking disagreements, redundancy density, bottleneck flags, and the failure-taxonomy rules in Appendix A.

Table 10 shows that the decomposition fields retrieve the rule-defined audit targets more effectively than the scalar-only view. Bottleneck protection, redundancy consolidation, weak utility-anchor repair, and structural over-correction are explicitly encoded in the SC-FMA fields. Table 11 illustrates how these fields change the review action attached to a ranked step.

4.6.5. Variant-Selection Policy

The stratified findings carry a practical recommendation for future KBS-facing planning at the methodology level. When intermediate verification traces are relatively short and a reasonable fidelity baseline already captures annotation ordering, as in PRM800K under $\mathbf{w}_{\text{struct}}$, Ridge is the recommended variant. It soft-averages structural information into the fidelity signal without aggressive weight reallocation. QP may be worth evaluating under controlled conditions when traces are long and redundancy density is high, as may happen in multi-hop KG reasoning or retrieval-augmented pipelines with overlapping intermediate supports. The current PRM800K evidence shows, however, that QP’s hard redundancy penalty can overshoot when learned features already encode structural information. On long traces, this produces lower ranking quality (Spearman $\rho = 0.172$). The policy in Table 12 is therefore an engineering guideline, not a theoretical guarantee.

Table 11

Compact Audit Card case study: scalar ranking versus SC-FMA decomposition.

Failure target	w_{struct} -only view	SC-FMA Audit Card view
Bottleneck protection	A scalar score can rank the step but does not identify that the selected step gates downstream reasoning.	Bottleneck and necessity fields mark the step as dependency-exposed, so the action is to protect or inspect the downstream chain before dropping it.
Redundancy consolidation	Multiple high-scored steps can look independently important.	Redundancy density and neighborhood fields identify a cluster, so the action is to consolidate the checks or sample the cluster rather than inspect every duplicate.
Weak utility anchor	A high learned score hides whether the raw local utility signal failed.	Fidelity and raw-anchor disagreement expose an upstream utility-anchor problem, so the action is to repair the signal source rather than blame graph structure.

Table 12

Ridge/QP selection policy derived from the stratified analysis. This is an engineering guideline, not a theoretical guarantee.

Scenario	Recommendation
Strong-fidelity signal, as in PRM800K-like process annotations	Ridge
High structural conflict or weak-fidelity signal	QP, validated on development data first

The variant-selection policy should be calibrated on a development set before use. It should not be treated as an optimal decision rule across new task distributions.

4.6.6. Practical Implications

The results illustrate a recurring KBS design tension. Local evidence signals, such as retrieval confidence scores, entity-linking similarity, and rule-certainty factors, are often abundant but poorly aligned with audit needs. A high-confidence retrieval may duplicate an earlier check, while a low-confidence entity disambiguation may gate downstream inference. SC-FMA makes this tension visible by linking the queue to redundancy and dependency fields rather than to a scalar score alone.

For KBS review, the practical implication is an audit interpretation policy. High-necessity steps become review priorities, high-redundancy steps become consolidation candidates, bottleneck steps are surfaced for conservative retention, and weak utility-anchor cases indicate that the scoring signal should be inspected before blaming graph structure. This gives the reviewer a vocabulary for acting on a queue: follow local evidence, consolidate repeated checks, protect dependency-exposed steps, or repair the fidelity signal.

5. Conclusions

This paper presented SC-FMA, a structurally calibrated method for turning a supplied utility or proxy-fidelity signal into an inspectable audit queue. Its SCU objective combines fidelity, graph necessity, redundancy, and bottleneck terms so that a ranked queue is accompanied by the reason each step is prioritized. The central contribution is this decomposition: it preserves useful ranking signal when the supplied anchor is strong, while making the structural role of each selected step inspectable, so that a reviewer can tell whether a high-ranked step is local evidence to follow, a redundant cluster to consolidate, a structural bottleneck to protect, or an upstream signal to repair.

The empirical evidence supports this role across the Evidence Ladder. On the controlled synthetic stage, SC-FMA QP captures designed structural preferences and lifts mean Spearman correlation from 0.483 for raw CIU to 0.597 (five-seed mean). On the PRM800K locked stage, SC-FMA Ridge closely preserves the learned w_{struct} fidelity signal ($\rho = 0.604$ versus 0.611; NDCG@25% of 0.946 versus 0.951) while exposing the same decomposed fields, and the

oracle audit-target validation stage shows the decomposition lifts Recall@25% from 0.235 to 0.699 and NDCG@25% from 0.353 to 0.978 over a scalar-only view.

Beyond ranking metrics, the stratified analysis yields an actionable variant-selection policy rather than a single leaderboard number: Ridge is the robust choice when a learned fidelity signal already aligns with annotations, whereas QP is reserved for settings where utility and structure disagree. The decomposition also gives reviewers a vocabulary for acting on a queue — follow local evidence, consolidate repeated checks, protect dependency-exposed steps, or repair the fidelity signal — turning a ranked list into a budgeted audit plan.

Together, these results establish SC-FMA as a foundational calibration layer for audit prioritization in knowledge-intensive reasoning: it preserves ranking accuracy while making the structural reason behind each priority inspectable. The current evidence is strongest for synthetic calibration and PRM800K-like step-label ranking; extending SC-FMA to deployment or training claims is left as future work, including human-centric evaluation of reviewer efficiency, typed ontology integration for domain-specific graph backends, and validation on live KBS maintenance traces.

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Declaration of Competing Interest

The authors declare that they have no conflicts of interest related to this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI GPT-5 to improve language clarity and assist with LaTeX formatting checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

PRM800K, MuSiQue, and WebQSP are publicly available from their original sources. Derived locked-split reports, audit-prioritization artifacts, trace-audit diagnostics, experiment configurations, and reproduction scripts will be deposited in an anonymous public repository for review and released with the final article. Full commands, local raw-data expectations, output paths, and claim-boundary notes are documented in the supplementary reproducibility checklist.

CRedit authorship contribution statement

Haoran Ma: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Ningning Wang:** Methodology, Validation, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

A. Failure Taxonomy Appendix

This appendix provides diagnostic support for selecting variants in PRM800K-like audit prioritization. It stays within the real-data audit-prioritization boundary used in the main text.

Each Audit Card’s Decision Label maps to one of three canonical actions derived from the SCU decomposition and unavailable from the fidelity-only scalar: protect a bottleneck, consolidate a redundant cluster, or repair an upstream utility anchor.

Table 13

Reviewer V2 failure taxonomy distribution on the locked PRM800K split (4,417 traces). Labels are multi-label, so percentages need not sum to 100%.

Label	Count	Percentage
structural_over_correction	2145	48.56%
redundancy_misclassification	1214	27.48%
bottleneck_over_protection	382	8.65%
weak_utility_anchor	2420	54.79%
low_signal_or_tie	809	18.32%

Redundancy misclassification and bottleneck over-protection appear often enough to discuss directly as variant selection guidance. The following Audit Cards use a fixed template so that each example links the ranking change to a concrete reviewer action. They provide explainability evidence for how decomposition turns a flat ranking disagreement into an audit task; they are not a substitute for a full user study.

Table 14

Audit Card 1: bottleneck-protected structural over-correction.

Field	Audit Card
Trace	prm800k_pool_010077_26cf715105; 7 steps. Representative step: “Now, I wonder if there is a geometric interpretation of the problem.”
Observed issue	Labels include structural_over_correction and bottleneck_over_protection. Step 3 has a below-maximum label (0.5) but receives the largest QP weight (0.2493).
Direct ranking behavior	w_{struct} keeps the high-label early vector-setup steps near the top (0.8966–0.9510) and gives the final false conclusion almost no weight (0.0027).
SC-FMA decomposition	Fidelity: w_{struct} keeps the high-label setup steps high. Necessity: graph exposure marks the reinterpretation as structurally visible. Redundancy: little consolidation is needed in this short trace. Bottleneck: QP protects the lower-label step, creating the audit disagreement.
Why not w_{struct} alone?	w_{struct} gives the right ordering but not the reason a lower-label step became structurally protected. SC-FMA exposes that the disagreement is bottleneck-driven, not generic ranker behavior.
Reviewer Decision	Inspect the bottleneck-protected lower-label step before accepting the structural floor; check whether the geometric reinterpretation is a true downstream dependency or an over-protected detour.
Decision Label	PROTECT_BOTTLENECK
Claim boundary	Diagnostic PRM800K Audit Card only; it supports variant selection and reviewer triage, not task-success validation.

Table 15

Audit Card 2: redundancy over-merging in a long trace.

Field	Audit Card
Trace	prm800k_pool_012035_e6055a1cb2; 30 steps. Representative step: “One way to do that is to pick a random value of x and find the corresponding value of $y = f(x)$, then swap the coordinates.”
Observed issue	Labels include <code>structural_over_correction</code> , <code>redundancy_misclassification</code> , and <code>bottleneck_over_protection</code> . Redundancy density is high (0.5749), and QP collapses several early high-label setup steps to nearly zero.
Direct ranking behavior	w_{struct} assigns consistently high scores to the graph-sketch, asymptote, intercept, and checking steps (roughly 0.91–0.97), while QP reallocates mass away from several of these high-label setup steps.
SC-FMA decomposition	Fidelity: w_{struct} keeps many graph-analysis checks high. Necessity: downstream exposure concentrates attention on a narrow band. Redundancy: similar graph checks are treated as overlapping. Bottleneck: protected steps compete with high-label setup steps for review budget.
Why not w_{struct} alone?	Direct ranking shows which steps are high, but it cannot tell whether the trace contains redundant audit units or whether several superficially similar graph checks should be consolidated. The decomposition makes that audit decision explicit.
Reviewer Decision	Check redundancy over-merging: decide whether asymptote identification, intercept checks, and coordinate-swap verification are separate audit units or can be reviewed as one cluster.
Decision Label	CONSOLIDATE_REDUNDANT_CLUSTER
Claim boundary	Diagnostic PRM800K Audit Card only; it motivates reviewer workflow design and does not validate live KBS traces.

Table 16

Audit Card 3: weak local-utility anchor.

Field	Audit Card
Trace	prm800k_pool_006217_43f4f279ef; 3 steps. Representative step: “I call this altitude $\overline{AF}$, where F is the midpoint of BC .”
Observed issue	Labels are [0.5, 1.0, 0.0]. Raw local utility is inverted: it gives the false parallel-line step the largest score (1.2159) and the highest-label midpoint-altitude step a lower score (0.7647).
Direct ranking behavior	w_{struct} , QP, and Ridge rank the midpoint-altitude step highest and the false parallel-line step lowest; raw local utility has Spearman $\rho = -1.0$ on this trace.
SC-FMA decomposition	Fidelity: the learned signal corrects the raw utility inversion. Necessity: the altitude construction remains the structurally meaningful check. Redundancy: no consolidation decision is needed in the three-step trace. Bottleneck: no bottleneck flag is needed, so the reviewer should repair the utility anchor rather than the structure detector.
Why not w_{struct} alone?	w_{struct} fixes the local ordering, but the decomposition identifies the failure mode as a weak utility anchor rather than redundancy or bottleneck pressure. That distinction tells the reviewer which upstream signal needs repair.
Reviewer Decision	Inspect why the local utility anchor rewarded the false parallel-line step; use the structured decomposition for triage and avoid treating raw utility as an independent audit target.
Decision Label	REPAIR_UPSTREAM_UTILITY
Claim boundary	Diagnostic PRM800K Audit Card only; it supports signal-quality auditing, not external reasoning-task generalization.

B. Runtime and Reproducibility

Runtime and reproducibility metadata are summarized in Table 17. The records below are wall-clock measurements from the reproducibility summary; API calls remain zero and frozen artifacts are read-only inputs.

Table 17

Runtime and reproducibility summary for the experiments. Wall time is measured in seconds; API calls are zero for all runs.

Route	Samples	Steps	Seeds	Elapsed (s)	Hardware	API Calls
SCU Component Contribution	200	1,008	5	30.7961	CPU (12 cores)	zero
Graph Construction Ablation	800	2,400	5	3.4197	CPU (12 cores)	zero
Failure Taxonomy	4,417	34,219	N/A	88.6221	CPU (12 cores)	zero

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